



Food Poisoning — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. According to the WHO, roughly how many deaths does unsafe food cause globally each year?

EASY

- A. 15,000
- B. 150,000
- C. 15 million
- D. 1.5 million
- E. 150 million

2. What are the three links of the food poisoning chain?

EASY

- A. Bacteria, sunlight, and low humidity
- B. Toxins, refrigeration, and acidity
- C. Viruses, high pH, and cold storage
- D. Bacteria on food, the right conditions to grow, and time to multiply
- E. Fungi, oxygen, and darkness

3. In food intoxication, symptoms typically appear within:

EASY

- A. 24–48 hours
- B. 2–12 hours
- C. 1 week
- D. 1 month
- E. Immediately, within seconds

4. Which organism is associated with the most lethal toxin known, with a calculated adult lethal dose of about 10 µg?

EASY

- A. *Clostridium botulinum*
- B. *Staphylococcus aureus*
- C. *Salmonella*
- D. *Vibrio cholerae*
- E. *Bacillus cereus*

5. Which toxin type explains the rapid vomiting seen in staphylococcal food poisoning?

MEDIUM

- A. Heat-stable preformed enterotoxin binding gut receptors and stimulating medullary centres
- B. Heat-labile toxin activating adenylate cyclase
- C. LPS causing systemic fever
- D. Neurotoxin blocking acetylcholine release
- E. Collagenase causing tissue necrosis

6. Which strains of *Clostridium perfringens* mainly cause food poisoning, and via which toxins?

MEDIUM

- A. Type A strains, via kappa and lambda toxins
- B. Type A strains, via alpha and theta toxins
- C. Type C strains, via beta toxin
- D. Type D strains, via epsilon toxin
- E. Type B strains, via iota toxin

7. Botulinum toxin acts by:

MEDIUM

- A. Activating the sympathetic nervous system
- B. Stimulating serotonin release in the gut
- C. Blocking acetylcholine release at the presynaptic neuromuscular junction
- D. Increasing vascular permeability directly
- E. Degrading intestinal mucin

8. An infant under 6 months develops constipation, weak cry and poor feeding. Which mechanism of botulism is most likely?

MEDIUM

- A. Ingestion of preformed toxin in honey only
- B. A congenital neuromuscular disorder
- C. Infection of the infant gut by *C. botulinum* with in vivo toxin production
- D. Maternal transmission of antibodies
- E. Vaccine-associated paralysis

9. Which pair correctly matches *Bacillus cereus* toxin forms to their features?

MEDIUM

- A. Emetic: heat-stable, short incubation, rice-associated; Diarrheal: heat-labile, longer incubation, meat/milk-associated
- B. Emetic: heat-labile, rice-associated; Diarrheal: heat-stable, meat-associated
- C. Both forms are heat-stable and rice-associated
- D. Both forms are heat-labile with identical incubation
- E. Emetic: 24-hour incubation; Diarrheal: 1-hour incubation

10. Which feature allows *Salmonella* to persist in the environment despite pasteurisation killing it in milk?

MEDIUM

- A. Resistance to freezing but not chilling
- B. Survival in moist earth for up to a year and dry earth for 16 months
- C. Complete destruction by pickling solutions
- D. Inability to survive outside a host
- E. Sensitivity to all disinfectants

11. *Salmonella enteritidis* is notable because it can be found:

MEDIUM

- A. Only on the eggshell surface
- B. Exclusively in raw beef
- C. Only in unpasteurised milk
- D. Inside the yolk of the egg
- E. Only in seafood

12. Which step in *Salmonella* pathophysiology directly causes fever?

MEDIUM

- A. Release of LPS
- B. Direct toxin binding to the vomiting centre
- C. Formation of gas gangrene
- D. Blockade of acetylcholine release
- E. Loss of skin turgor

13. Which clinical feature helps distinguish *Campylobacter jejuni* infection?

MEDIUM

- A. Projectile vomiting with minimal diarrhea
- B. Ascending paralysis
- C. Rice-water stools
- D. Jaundice with hepatomegaly
- E. Fever followed by foul-smelling, watery diarrhea, sometimes with blood and mucus, plus backache

14. Traveller's diarrhea is most commonly caused by which organism?

MEDIUM

- A. Enterohemorrhagic *E. coli* (EHEC)
- B. *Vibrio cholerae*
- C. *Clostridium botulinum*
- D. *Bacillus cereus*
- E. Enterotoxigenic *E. coli* (ETEC)

15. Cholera toxin causes diarrhea mainly by:

HARD

- A. Directly invading and destroying colonic epithelium
- B. Blocking neuromuscular transmission
- C. Producing a heat-stable emetic toxin
- D. Forming granulomas in the ileum
- E. Stimulating persistent secretion of isotonic fluid from gut mucosal cells

16. A patient with cholera produces 25 watery stools in a day and dies within 18 hours. Which mechanism best explains the rapid death?

HARD

- A. Direct toxin action on the heart muscle
- B. Massive fluid and electrolyte loss causing hypovolaemic shock
- C. Ascending paralysis of respiratory muscles
- D. Acute hepatic necrosis
- E. Anaphylactic shock from the toxin

17. A batch of maize stored in humid conditions caused an outbreak of fatal liver disease. Which agent and mechanism best fit?

HARD

- A. Botulinum toxin causing flaccid paralysis
- B. Cholera toxin causing secretory diarrhea
- C. Staphylococcal enterotoxin causing vomiting
- D. Shiga toxin causing haemolytic uraemic syndrome
- E. Aflatoxin from *Aspergillus* species causing acute hepatic necrosis

18. Comparing preformed-toxin food poisoning organisms, which two both classically involve heat-stable toxins not destroyed by boiling?

HARD

- A. *Clostridium perfringens* and the diarrheal form of *Bacillus cereus*
- B. *Salmonella* and *Campylobacter jejuni*
- C. *Staphylococcus aureus* and the emetic form of *Bacillus cereus*
- D. *Vibrio cholerae* and *E. coli* O157:H7
- E. *Clostridium botulinum* and *Clostridium perfringens*

19. Why is single-case identification of foodborne illness usually difficult, except in a condition like botulism?

HARD

- A. Because botulism always occurs in outbreaks
- B. Because stool culture is impossible in botulism
- C. Because botulism has no incubation period
- D. Because botulism has a distinctive clinical syndrome (descending paralysis), unlike most other foodborne illnesses
- E. Because single cases are never infectious

20. An outbreak affects several people who shared a buffet meal 48 hours earlier, with fever, cramps and diarrhea; eggs and poultry salad were served unrefrigerated for hours. Which organism fits best, and by which pathogenic class?

HARD

- A. *Staphylococcus aureus*, by preformed toxin ingestion
- B. *Vibrio cholerae*, by noninvasive toxin secretion
- C. *Salmonella*, by intracellular invasion of intestinal epithelium
- D. *Clostridium botulinum*, by preformed neurotoxin
- E. *E. coli* ETEC, by noninvasive toxin secretion

✓ Answer key

Q	Answer	Difficulty	Why
1	D	EASY	WHO estimates 866 million illnesses and 1.5 million deaths yearly from unsafe food.
2	D	EASY	Breaking any one link prevents food poisoning.
3	B	EASY	Intoxication (preformed toxin) acts faster than infection (≥ 24 h).
4	A	EASY	Botulinum toxin is described as one of the most lethal poisons known.
5	A	MEDIUM	Staph enterotoxins are heat-stable, bind a bowel receptor and trigger the vomiting centre.
6	B	MEDIUM	Food poisoning is mainly type A strains producing alpha and theta toxins.
7	C	MEDIUM	It inhibits ACh release, causing flaccid paralysis via the parasympathetic and motor systems.
8	C	MEDIUM	Infant botulism results from gut colonisation and in vivo toxin production, unlike classic foodborne botulism.
9	A	MEDIUM	The lecture separates the two forms exactly this way.
10	B	MEDIUM	Salmonella persists long-term in soil despite being killed by pasteurisation.
11	D	MEDIUM	The lecture specifically notes <i>S. enteritidis</i> inside the egg yolk.
12	A	MEDIUM	LPS release triggers fever in Salmonella infection.
13	E	MEDIUM	This combination, including backache, is characteristic of <i>C. jejuni</i> .
14	E	MEDIUM	ETEC is specifically named as the main cause of traveller's diarrhea.
15	E	HARD	<i>V. cholerae</i> 's potent enterotoxin drives massive isotonic fluid secretion.
16	B	HARD	Death in cholera results from rapid, extreme dehydration and hypovolaemic shock, not direct organ toxicity.
17	E	HARD	Poor storage promotes mould growth and aflatoxin production, which is hepatotoxic.
18	C	HARD	Staph enterotoxin and <i>B. cereus</i> emetic toxin are both heat-stable; <i>C. perfringens</i> and <i>B. cereus</i> diarrheal toxin are heat-labile.
19	D	HARD	The distinctive neurological syndrome of botulism allows recognition even as an isolated case.
20	C	HARD	A 48-hour incubation with fever and diarrhea after eggs/poultry fits Salmonella's invasive mechanism, distinct from the rapid preformed-toxin illnesses.